3.1H AND THE START OF 3.1I · PP. 72–74

Data records part of a situation

- A click records an action
- A system may infer a preference from that action
- The inference remains open to other explanations



AI conceptual illustration · not an actual company, venue or measured outcome

Big data and data analytics

- Big data involves large, varied collections of records
- Analytics processes data to find patterns and trends
- A useful result depends on the question and the quality of the data

Volume: amount	Velocity: arrival/processing rate
Variety: forms	Veracity: trustworthiness

A useful dataset fits the question; more records alone cannot repair missing context or systematic omission.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 72–73.

Volume and velocity

- Volume describes the amount of data
- Velocity describes how quickly data arrives and is processed
- Fast incoming data can create processing and checking demands

Situation	Characteristic to examine
A ten-year archive	Volume
Thousands of new events each second	Velocity
A daily batch analysed overnight	Arrival and processing delay

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Variety and veracity

> Variety concerns formats and types of data

> Veracity concerns accuracy, reliability and quality

> A larger collection can contain more of the same error

Small and clean, or large and messy?

- Dataset A: 40 verified records from one class
- Dataset B: 40,000 records with duplicates and missing fields
- Neither size alone nor cleanliness alone answers the question

Question	A	B
Known quality	Verified values	Needs checks
Coverage	One class	Unknown representation
Uncertainty	Generalization	Errors and selection

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73–74.

A missing value is an unresolved question

- Blank duration does not mean zero duration
- Dropping a record changes who remains in the dataset
- Imputing a value introduces an assumption

8 ÷ 1 = 8

Recorded values only

Blank is excluded and declared.

8 ÷ 2 = 4

Blank assumed zero

A different assumption changes the result.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73–74.

An association does not establish a cause

- A pattern may be useful for prediction
- Other factors may explain the relationship
- A causal conclusion needs an appropriate investigation

1. Observed association →

2. Alternative explanation →

3. Comparison evidence →

4. Qualified causal claim

A possible explanation requires testing. Correlation alone cannot identify intention or establish cause.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73.

APPLY · 8 · TEXTBOOK PP. 72, 74

The data lifecycle creates decisions at every stage

> Create or collect data

> Store and use it

> Preserve or destroy it when appropriate

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 72, 74.

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Collection can shape the eventual result

- Collect data that fits a defined purpose
- Examine whose records are absent or overrepresented
- Consider permission and unnecessary collection

1. Defined purpose →

2. Choose relevant fields →

3. Check omitted groups →

4. Decide retention

Collection choices shape both evidence and privacy costs.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 74.

Analytics can support several purposes

- Describe a past pattern
- Detect an unusual record or event
- Predict a possible future outcome

A bounded big-data inquiry

> Question: whose experience does this dataset describe?

> Check collection, formats, missing records and quality

> Match each claim to evidence that can support it

Chapter coverage and delayed retrieval

> This deck teaches 3.1H and collection at the start of 3.1I

> Earlier 3.1 sections and later lifecycle dilemmas remain separate

> Explain the four Vs in a new context without notes

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 72–74.